Product Requirement Document (PRD)

Project Name: AI-Powered Travel Planner

1. Overview

The AI-powered Travel Planner is an intelligent mobile application designed to help users plan personalized trips efficiently. By leveraging AI, the app analyzes user preferences such as destination, budget, travel dates, and interests to generate an optimized, interactive itinerary. The goal is to simplify trip planning by providing smart recommendations, budget tracking, and seamless booking integrations, enhancing the overall travel experience.

2. Goals & Objectives

Product Goals:

- Apply fundamental UI/UX principles to ensure a smooth, intuitive, and enjoyable travel planning experience.
- Provide users with an interactive, AI-driven trip planning tool that is efficient and easy to use.
- Introduce AI-powered automation to personalize itineraries, reducing the time and effort spent on manual trip research.

User Experience Objectives:

- Ensure seamless navigation through user research, wireframing, and prototyping.
- Provide intelligent, real-time travel suggestions based on historical data, weather conditions, and user behavior.
- Enable collaborative trip planning with friends and family for shared travel experiences.

3. Features & Functionality:

User Onboarding & Profile Management;

- Signup/Login – Users can register via Google, Apple ID, or email for a personalized experience.
- Profile Creation – Users can set up a profile including name, preferred travel styles, past trips, and favorite destinations.
- Data Synchronization – Users' preferences are stored securely, enabling AI-driven personalization.

Trip Planning & AI-Powered Recommendations;

- User Input Form – Users provide essential trip details such as destination, budget, travel dates, and interests.

- AI-Powered Itinerary Generator – AI processes the input and generates an optimized day-by-day trip plan.
- Accommodation & Transport Suggestions – The AI recommends hotels, flights, local transport, and rental options.
- Alternative Itinerary Suggestions – If a user's budget or schedule changes, AI offers adaptable plans.

Interactive Travel Itinerary;

- Visual Itinerary Display – A day-by-day breakdown of activities, showing attractions, restaurants, and transport details.
- Map Integration – Users can view locations, travel routes, and estimated travel time between activities.
- Editable Itinerary – Users can add, remove, or rearrange activities to customize their trip.

AI Enhancements & Personalization;

- Smart Recommendations – AI continuously learns user preferences and refines future itinerary suggestions.
- Weather-Based Activity Suggestions – AI adapts plans based on real-time weather conditions.
- AI-Generated Travel Tips & Packing Lists – Users receive personalized checklists based on their destination and travel style.
- Local Event & Festival Recommendations – AI suggests events happening during the user's visit.

Budget & Expense Tracking;

- Trip Cost Breakdown – AI estimates total trip costs based on itinerary selections.
- Expense Tracker – Users can manually log expenses to track their spending during the trip.

Collaboration & Social Sharing;

- Trip Sharing – Users can share itineraries with family and friends via social media or messaging apps.
- Collaborative Trip Planning – Multiple users can edit and contribute to the same itinerary, making group trips easier to plan.

Booking & Third-Party Integrations;

- Hotel, Flight & Activity Booking – The app integrates with popular booking platforms (e.g., Booking.com, Expedia, Airbnb).
- In-App Links for Bookings – Users can seamlessly book flights, hotels, and activities from within the app.

Notifications & Reminders;

- Travel Countdown – Users receive notifications leading up to their trip.
- Flight Check-in Alerts – AI reminds users to check in for their flights on time.

- Activity Reminders – Users receive real-time reminders for scheduled activities.

Offline Mode;

- Downloadable Itinerary – Users can access their itinerary offline, ensuring they can navigate even without internet access.

User Feedback & Support;

- Ratings & Reviews – Users can rate and review suggested places and experiences.
- AI-Powered Chatbot – AI answers travel-related queries and assists users with itinerary changes.

4. Target Audience

- Frequent travelers who seek AI-driven, time-saving recommendations.
- Users who struggle with itinerary planning and want an automated solution.
- Budget-conscious travelers who want to track expenses efficiently.
- Group travelers who require a collaborative itinerary planning tool.

5. AI Integration & Technical Considerations

- Dynamic Itinerary Optimization – AI continuously refines travel plans based on real-time conditions.
- Historical Data Analysis – AI learns user behavior to provide more relevant recommendations over time.
- Context-Aware Smart Suggestions – AI recommends activities based on location, weather, and events.

AI Interaction Points with Users

AI Functionality	**User Interaction**
Personalized Travel Plan Generation	AI suggests custom itineraries based on user preferences
Dynamic Activity & Restaurant Sugge	AI recommends local attractions, dining, and experiences
Real-Time Optimization	AI updates schedules based on weather conditions and user feedback
Budget-Aware Planning	AI ensures plans stay within the user's budget

Technical Architecture;

- Backend Services – AI recommendation engine powered by machine learning models.
- Frontend UI/UX – User-friendly interface optimized for mobile responsiveness.
- API Integration – Connectivity with Google Maps, flight/hotel booking services, and payment gateways.
- Authentication – Secure Google, Apple ID, and email login for user accounts.